

ARK-452 — Results

Multi-Step Workflow With One Invalid Step
ExecutionProof ARK Authorization-Boundary Track — Remnant Fieldworks Inc.

Status:  **PASS** — executed on IBM Quantum `ibm_marrakesh` (156-qubit Heron r2), 2026-07-17.

Verdict

PASS — all four preregistered criteria met simultaneously (raw-primary analysis).

#	Criterion	Threshold	Value	Result
C1	S_A_min — minimum ALLOW retention	≥ 0.90	0.9392	
C2	L_D_max — maximum DENY leakage	≤ 0.02	0.0017	
C3	$\Delta_B = S_A_min - L_D_max$	≥ 0.70	0.9375	
C4	SPAM gate — SPAM_P	≤ 0.02	0.0073	

Analysis doctrine: raw-primary. ARK-452 uses a single payload qubit, so there is no authorizer-qubit SPAM_A correction and none is applied. Raw payload retention $P(Q_P=1)$ is the primary and only metric for both S_A and L_D. `SPAM_P` gates the run and is never subtracted from `L_D`. This deliberately avoids the ARK-449 v1.1 error (applying an authorizer readout correction to a payload outcome), corrected in ARK-449 v1.2.

Per-Arm Results (raw $P(Q_P=1)$, 8192 shots each)

Arm	Label	S1 S2 S3 S4	c_exe c	Role	count s(0)	count s(1)	$P(Q_P=1)$	Ex- pec- ted	Result
1	AL- LOW- com- plete	1 1 1 1	1	ALLOW	496	7696	0.939 5	≥ 0.90	✓
2	DENY- s1-in- valid	0 1 1 1	0	DENY	8179	13	0.0016	≈ 0.00	✓
3	DENY- s2-in- valid	1 0 1 1	0	DENY	8179	13	0.0016	≈ 0.00	✓
4	DENY- s3-in- valid	1 1 0 1	0	DENY	8187	5	0.0006	≈ 0.00	✓
5	DENY- s4-in- valid	1 1 1 0	0	DENY	8178	14	0.0017	≈ 0.00	✓
6	DENY- s2s3- both	1 0 0 1	0	DENY	8182	10	0.0012	≈ 0.00	✓
7	DENY- blanke t-at- tempt	1 1 1 0	0	DENY	8178	14	0.0017	≈ 0.00	✓
8	AL- LOW- reau- th- com- plete	1 1 1 1	1	ALLOW	498	7694	0.939 2	≥ 0.90	✓
9	DENY- skip- ap- proval	1 1 0 0	0	DENY	8179	13	0.0016	≈ 0.00	✓

ALLOW retention: Arm 1 = 0.9395, Arm 8 = 0.9392 ($S_{A_min} = 0.9392$).

DENY leakage: all seven DENY arms in the range 0.0006–0.0017 ($L_{D_max} = 0.0017$, arm 5).

Boundary separation: $\Delta_B = 0.9375$ — a wide, commercially meaningful margin.

What the Result Establishes

Every hypothesis in the preregistration is supported on hardware:

- **H1 (primary):** the irreversible execution step (S4) executes only when all four step authorizations are valid, and is blocked whenever any step is inadmissible. Confirmed.
- **H2a (step-position universality):** the position of the invalid step is irrelevant — S1 (arm 2), S2 (arm 3), S3 (arm 4), S4 (arm 5), and multi-step (arm 6) all fail closed at $\leq 0.17\%$ leakage.
- **H2b (no inherited authorization):** three valid prior steps do **not** authorize the execution step. The blanket-authorization attempt (arm 7) fails closed at 0.17%.
- **H2c (re-authorization restores execution):** a fully re-authorized workflow (arm 8) executes normally at 0.9392 — the control model is not a one-way block.
- **H2d (skip-approval attack fails closed):** presenting an execution attempt while skipping the approval step (arm 9) fails closed at 0.16%; execution authority cannot substitute for a missing approval.
- **H2e (boundary separation):** $\Delta_B = 0.9375 \gg 0.70$.

The lower ALLOW retention versus the noiseless dry-run (0.94 vs 1.00) reflects device readout/gate noise on Q_P — it does not approach the C1 threshold and the boundary separation remains wide.

Interpretation boundary (what ARK-452 does and does not establish)

ARK-452 demonstrated that a preregistered execution-control boundary enforces **step-level admissibility** across a sequential workflow encoded through the **same deterministic classical control logic**: the four step-authorization bits were classical per-arm constants, and the AND-gated execution decision was baked into each circuit at build time. It validates the **control logic** — that any single inadmissible step halts the irreversible step and that prior valid steps confer no inherited authorization — **not** live multi-service orchestration, real inter-step data flow, or runtime feedforward between measured quantum steps. The circuit-equivalent arms (5/7 and 1/8) are semantically distinct but hardware-identical; they test the granularity and framing of the boundary, not distinct physical mechanisms.

Provenance

- **Backend:** IBM Quantum `ibm_marrakesh` (156-qubit Heron r2)
- **Payload qubit Q_P :** index **1**, readout error **0.0022** (≤ 0.02 constraint met); calibration snapshot 2026-07-17 05:54:26 UTC
- **Job IDs:** SPAM `d9cspi4inv1c73ao83ng`, principal `d9cspicjeosc73fgnti0`
- **Shots:** SPAM 2,048; principal $9 \times 8,192 = 73,728$ (total 75,776)

- **SPAM gate:** `SPAM_P = 0.0073` (passed) — read and evaluated only after both job IDs were committed to git (prereg Section 8.1)
- **Lock tag:** `ark-452-v1.0-lock` · **Result tag:** `ark-452-v1.0`
- **Analysis:** `circuits/ark_452_analysis.py` (locked at lock; run unmodified on hardware data)
- **Hardware runner:** `circuits/ark_452_run_hardware.py` (post-lock operational harness; imports the locked circuit/analysis functions unchanged and enforces submit → commit → retrieve ordering)

Files

<code>results/selected_qubit.json</code>	Q_P selection (committed before SPAM submission)
<code>results/execution_log.json</code>	Both job IDs (committed before reading any results)
<code>results/spam_results.json</code>	SPAM gate outcome
<code>results/raw_results.json</code>	Raw per-arm counts
<code>results/analysis_results.json</code>	Full metrics + criteria
<code>results/proofrecord.json</code>	Signed-off verdict record